

The Components Manifesto

We are not building another electronics editor.

There are already excellent tools for drawing schematics.

There are already excellent tools for PCB design.

There are already excellent hardware description languages.

There are already excellent simulators.

Components is not trying to replace any of them.

Components exists for a different reason.

We believe that understanding comes before implementation.

Many tools teach people how to use software.

We want to help people understand electronics.

A student should understand why a circuit works before worrying about menus, syntax, or file formats.

A circuit should remain the same regardless of how it is viewed, edited, simulated, or exported.

Meaning comes first.

We separate meaning from presentation.

A circuit has meaning.

A diagram is only one way to see it.

A breadboard is another.

A timing diagram is another.

A Verilog module is another.

A voice conversation is another.

An AI explanation is another.

All are representations of the same circuit.

The meaning exists only once.

Everything else is a projection.

We believe electronics should be conversational.

People should be able to build circuits by:

writing,

drawing,

dragging,

speaking,

or collaborating with AI.

Every interaction should ultimately describe the same Component.

Humans and AI should speak the same language.

We believe in construction before coding.

Components is inspired by learning through construction.

A learner does not begin by writing hardware description languages.

A learner begins by building.

By connecting.

By observing.

By asking questions.

By making mistakes safely.

The language should support this process.

The platform should never hide it.

We believe software should disappear.

The goal is not to learn Components.

The goal is to learn electronics.

The interface should become almost invisible.

The learner should think about registers,

not ribbons.

About buses,

not dialog boxes.

About ideas,

not software.

We believe every idea has one owner.

Circuit meaning belongs to the Component.

Presentation belongs to the Board.

Dependencies belong to Resources.

Simulation belongs to the Platform.

Natural language belongs to the AI Router.

No concept should exist twice.

No representation should redefine another.

We believe platforms should outlive technology.

Programming languages change.

Operating systems change.

AI models change.

User interfaces change.

Hardware changes.

The architecture should not.

Components is designed to survive its own implementations.

We believe AI is a participant, not a feature.

AI is not an optional button.

AI is another client of the platform.

A teacher,

a student,

a command-line tool,

a Board,

and an AI assistant

all communicate through the same model.

No client receives special treatment.

We believe openness creates longevity.

Boards should be replaceable.

AI providers should be replaceable.

Storage systems should be replaceable.

Programming languages should be replaceable.

Renderers should be replaceable.

Only the meaning of the circuit should remain stable.

We believe learning should remain simple.

A child should be able to build a circuit.

A teacher should be able to explain it.

An engineer should be able to verify it.

An AI should be able to understand it.

The same project should support all four.

We are building a platform, not an application.

Applications come and go.

Platforms create ecosystems.

Components exist so that others may build:

Boards.

Libraries.

Providers.

AI assistants.

Teaching systems.

Research tools.

Future ideas we cannot yet imagine.

This is our promise.

We will protect the meaning of the circuit.

We will keep architecture above implementation.

We will prefer simplicity over cleverness.

We will choose openness over lock-in.

We will design for decades, not for releases.

Technology will evolve.

The model will endure.